

A top-down view of several children sitting around a large white sheet of paper on a table, drawing with colored pencils and markers. The paper is filled with various hand-drawn illustrations including school supplies like a ruler, compass, and calculator; household items like a clock, lightbulb, and backpack; and educational concepts like a globe, a number grid, and a multiplication problem. The children's hands and heads are visible at the edges of the frame, actively engaged in the drawing process.

FAMILY ENVIRONMENT AND FAMILY STRUCTURE IN SHAPING CHILDREN'S MENTAL WELL-BEING

A STUDY OF TYPICALLY DEVELOPING AND ASD CHILDREN IN URBAN PABNA

PABNA UNIVERSITY OF SCIENCE AND TECHNOLOGY

Department of Statistics

Course Title : Statistical Field Survey

Course Code : STAT-4110

PRESENTED BY:

1. K.M Sumaiya Akter (211617)
2. MD. Abu Hanif (201603)
3. MD. Rifat Hasan Shakil(211641)
4. Aditto Ahosan Kabbo(211624)
5. Joyaria Tasmim(211633)

SUPERVISOR:

Dr. Md. Shamim Reza
Professor , Department of Statistics
Pabna University of Science and Technology



OUR RESEARCH FOCUS IS:
WHETHER CHILDREN'S
MENTAL WELL-BEING
DEPENDS MORE ON FAMILY
TYPE OR QUALITY OF FAMILY
ENVIRONMENT?

RESEARCH PROBLEM

WHY THIS STUDY MATTERS?

Family is the first environment where children learn behavior, communication, emotion, and social interaction.

Family environment includes:

- Emotional safety
- Calmness at home
- Parent-child quality time
- Family support
- Learning encouragement

Family structure includes:

- Nuclear family
 - Joint family
 - Single-parent family
-

RESEARCH OBJECTIVES

To examine how **family environment** and **family structure** influence children's mental well-being among TD and ASD children in urban Pabna.

Family Background

- Describe the family environment of children in urban Pabna
- Identify major family structures: nuclear, joint, and single-parent families

Child Mental Development

- Assess children's mental well-being using parent-reported information
- Compare mental well-being between typically developing (TD) children and children with ASD

Relationship Analysis

- Examine the relationship between family environment and children's mental well-being
 - Compare mental well-being across different family structures
-

RESEARCH GAP & OUR CONTRIBUTION

STEP 1

What Exists

- Family and child development have been widely studied.
- Most studies are from Western countries
- Limited local research in Bangladesh
- ASD children are usually studied separately.



STEP 2

The Gap

- Few studies compare Typically Developing and ASD children together.
- Family type is studied more, but family care, support, and emotional safety are less emphasized.
- Screen exposure is rarely considered with family factors.



STEP 3

Our Study Fills It

What matters more for children's mental well-being -family structure or family environment?

STUDY DESIGN & POPULATION

A large crowd of diverse people, seen from an aerial perspective, is arranged to form the geographical outline of the world map. The individuals are dressed in a variety of colorful clothing, and their shadows are cast onto the light gray ground beneath them. The title 'STUDY DESIGN & POPULATION' is overlaid in white text on the left side of the image, with a short horizontal line above the word 'STUDY'.

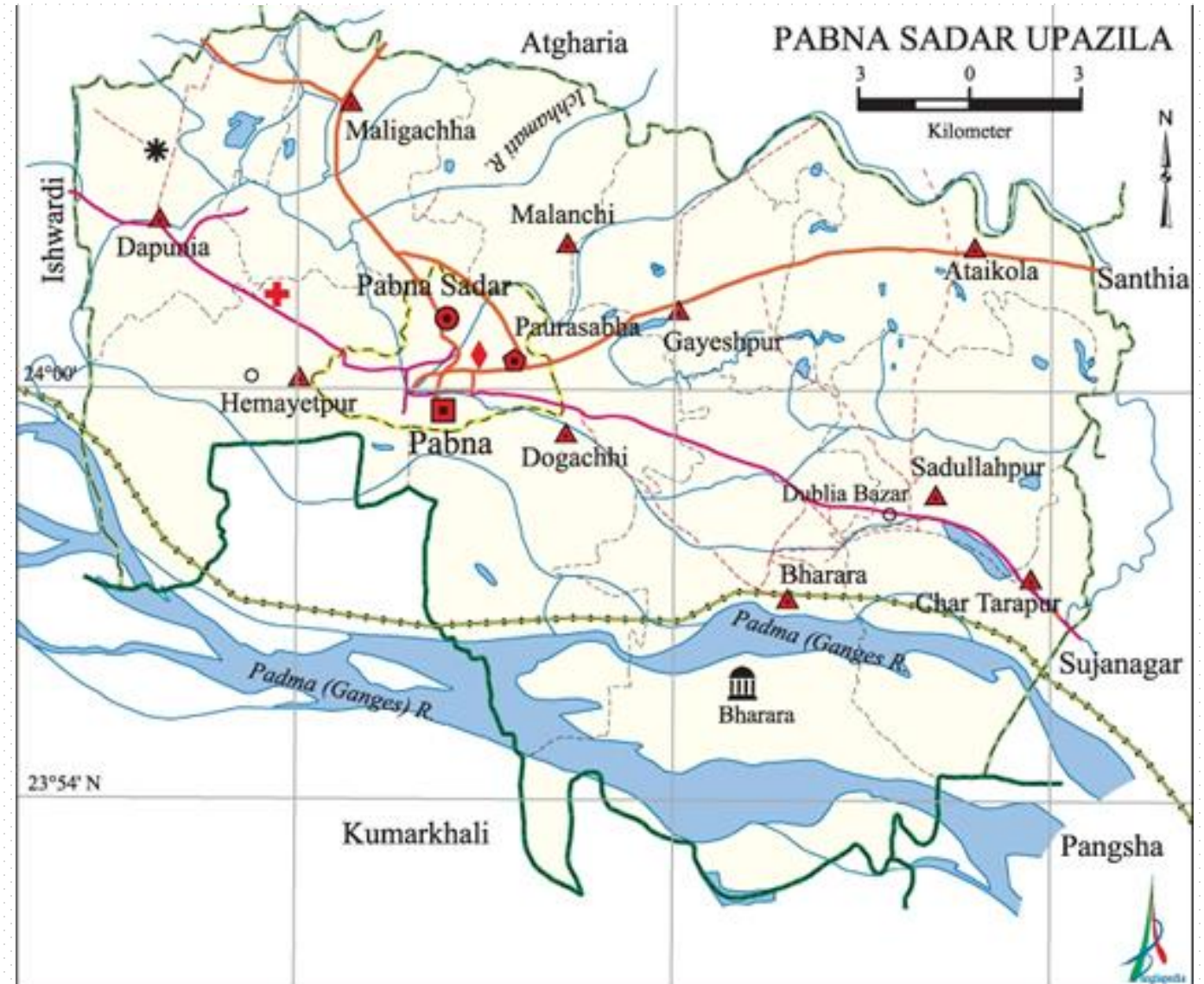
DESIGN

Study Design: Cross-sectional descriptive and analytical survey.

Study Area: Urban Pabna, Rajshahi Division, Bangladesh.

Study Population : Parents/caregivers of children aged **6–12 years**.

Child Groups: Typically developing and ASD children.



SAMPLE SIZE & SAMPLING

Estimated Sample Sizes

Method	ASD	TD	Total
Kelsey	71	213	284
Fleiss	76	228	304
Fleiss + Continuity Correction	85	254	339



Sampling Technique

Purposive sampling



Why Purposive Sampling?

ASD children were a specific target group and difficult to reach through random sampling.



Sample Size Determination

Sample size was calculated using two independent proportion comparison, because the study compared:

ASD children = Exposed group

Typically Developing children = Unexposed group



Key Assumptions:

Confidence level: 95%

Study power: 80%

TD : ASD ratio: 3 : 1

Expected prevalence: TD = 15% ASD = 30%

SAMPLE SIZE & SAMPLING



DATA COLLECTION, ENTRY & PROCESSING

Data Collection Tool :
Structured
questionnaire

**Data Collection
Method : Parent-
reported, paper-based
survey**



DATA COLLECTION, ENTRY & PROCESSING



Data Entry : using Google Forms/Sheets



Data Processing: Cleaning → Coding → Recoding → Score construction

STUDY VARIABLES & STATISTICAL ANALYSIS

Main Outcome : Mental Well-Being Score

Key Explanatory Variables : Family environment Score, family structure, child group

Other Factors : Screen exposure and socio-demographic variables

ANALYSIS FRAMEWORK

A. UNIVARIATE ANALYSIS

Described sample profile and variable distributions.

Output: Sample characteristics



B. SCALE VALIDATION

Reliability analysis was used to assess scale quality.

Output: Validated family environment and mental well-being scales



C. BIVARIATE ANALYSIS

Correlation, Chi-square, Mann-Whitney U, and Kruskal-Wallis tests were used to examine relationships and group differences.

Output: Group differences and associations



D. MULTIVARIATE ANALYSIS

The final analysis identified adjusted predictors of children's mental well-being.

Included steps:

- | | |
|----------------------------------|--------------------------------------|
| 1. Variable selection | 5. Regression diagnostics |
| 2. VIF / multicollinearity check | 6. Interaction / moderation analysis |
| 3. Multiple linear regression | 7. Supplementary logistic regression |
| 4. Hierarchical regression | 8. Machine learning comparison |

Output: Key adjusted predictors of mental well-being



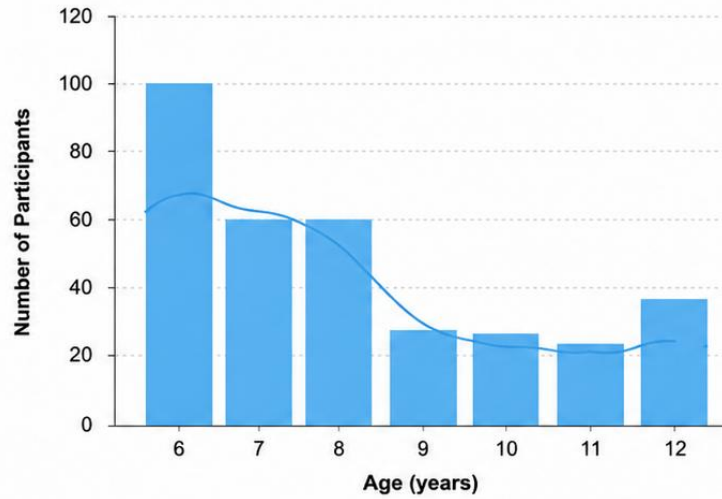
OVERALL GOAL

To identify and validate key factors associated with children's mental well-being through a rigorous, stepwise analytical approach.

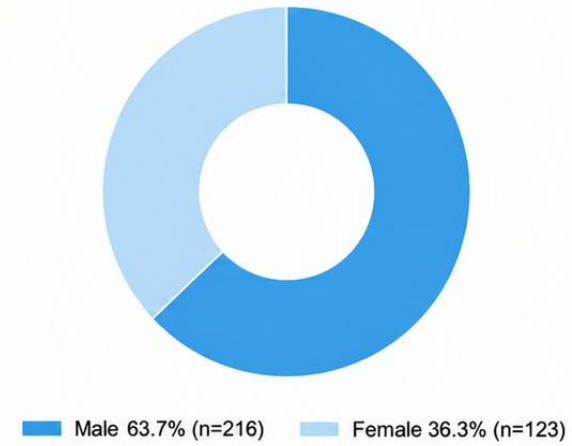
DESCRIPTIVE FINDINGS



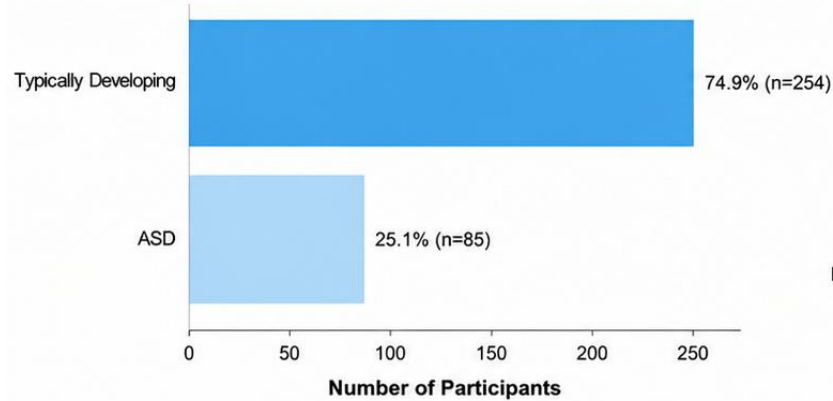
A | Child Age Distribution



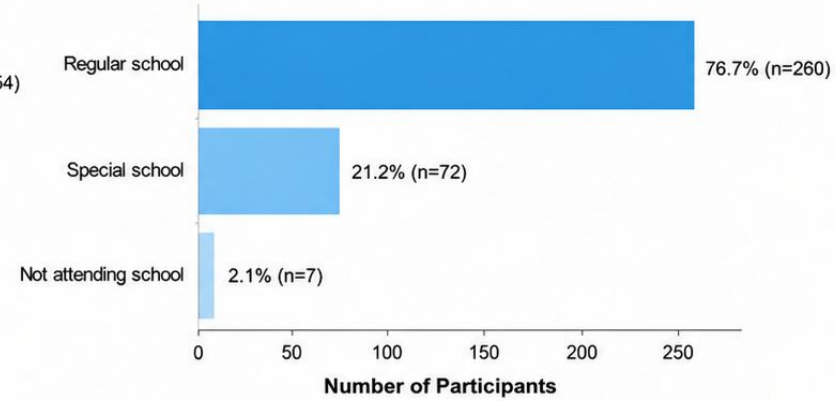
B | Child Gender Distribution



C | Child Group Distribution



D | School Status Distribution



CHILDREN'S DEMOGRAPHIC PROFILE

FAMILY ENVIRONMENT SCORE & MENTAL WELL-BEING SCORE

Mean Family Environment Score by family structure

Family Type	Mean	SD	N
Joint Family	3.624	0.621	108
Nuclear Family	3.682	0.593	217
Single-Parent Family	3	0.748	14

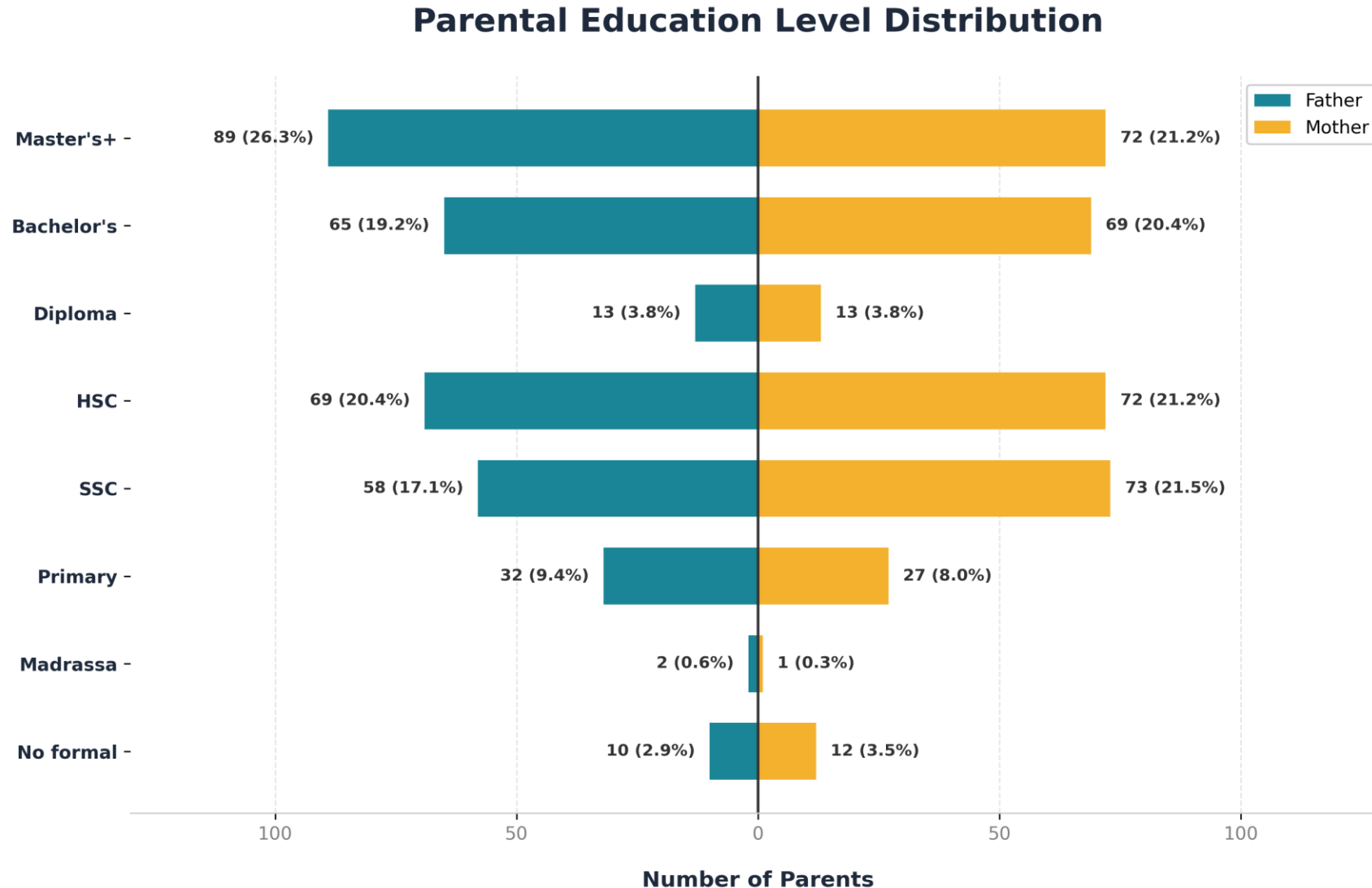
Interpretation:
Nuclear families had the highest mean family environment score, while single-parent families showed the lowest score.

Interpretation:
Typically developing children had a much higher mean mental development score compared to children with ASD.

Child Group	Mean	SD	Min	Max	N
ASD	2.482	0.74	1	4.5	85
Typically Developing	3.998	0.582	2	5	254

Mental Development Score by child group

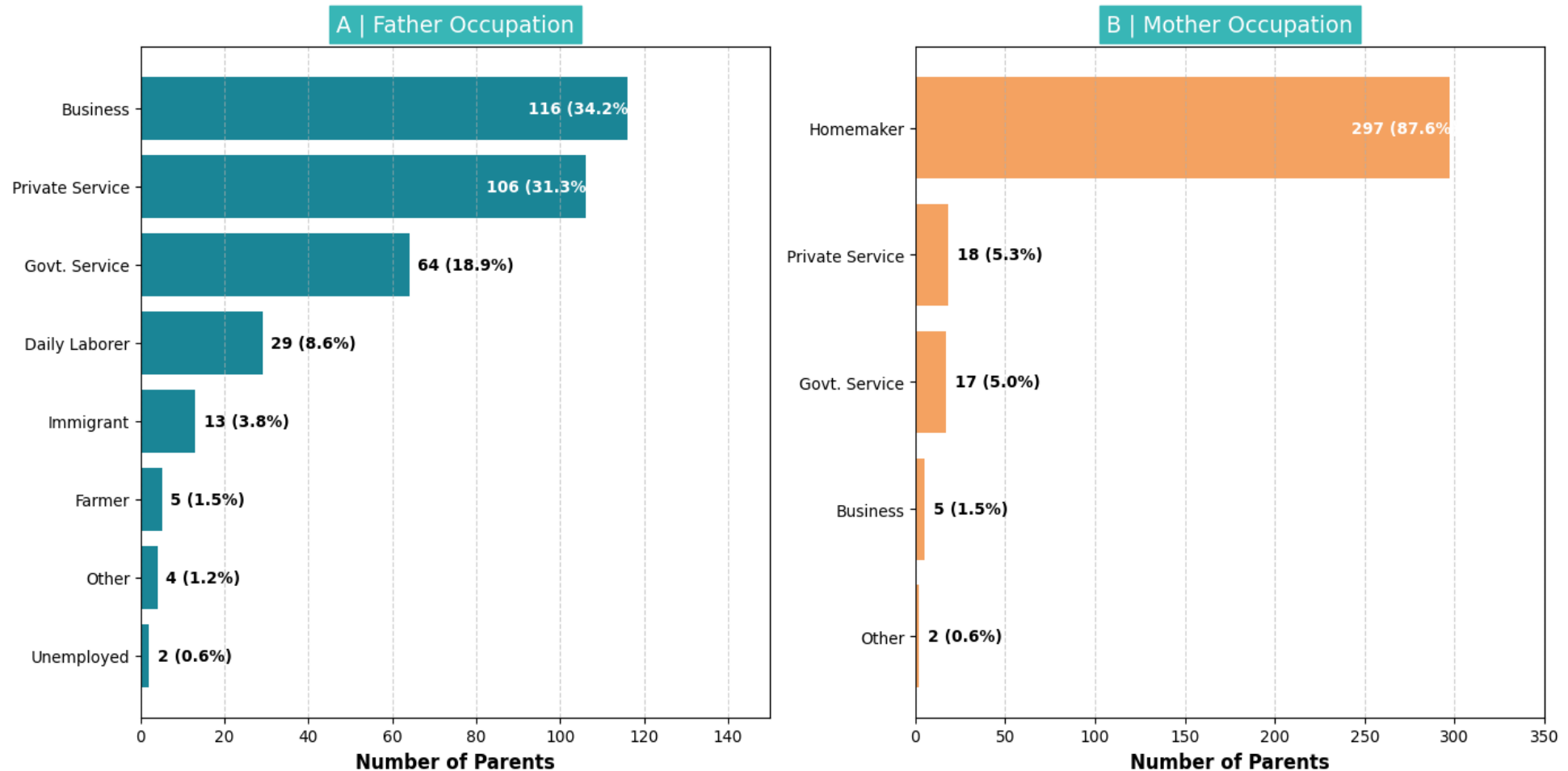
PARENTAL EDUCATION AND PARENTAL OCCUPATION



Note: Shows the highest level of education attained by the parents (N=339).

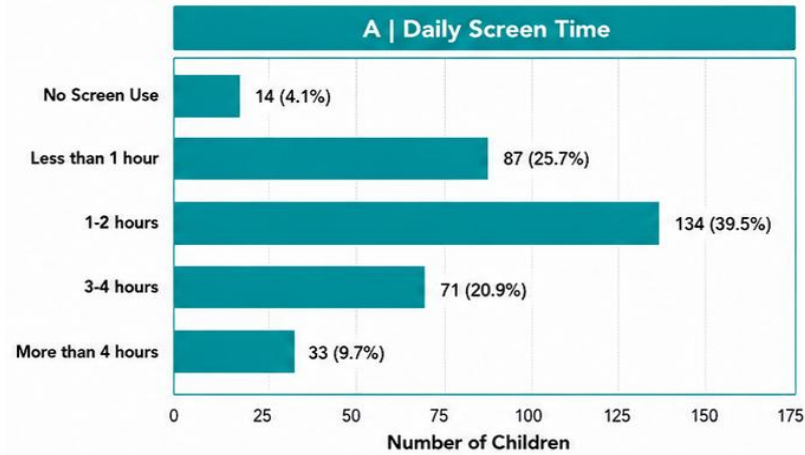
PARENTAL EDUCATION AND PARENTAL OCCUPATION

Parental Occupation Profile

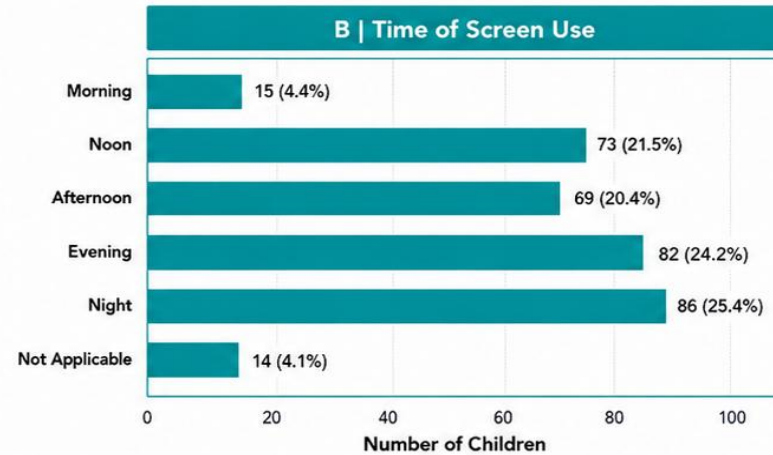


Note: Shows the distribution of primary occupations among fathers and mothers (N=339).

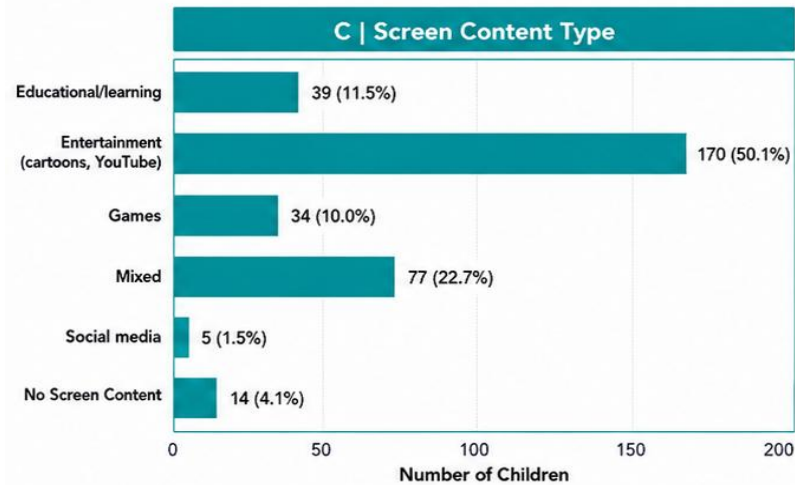
Screen Exposure Profile



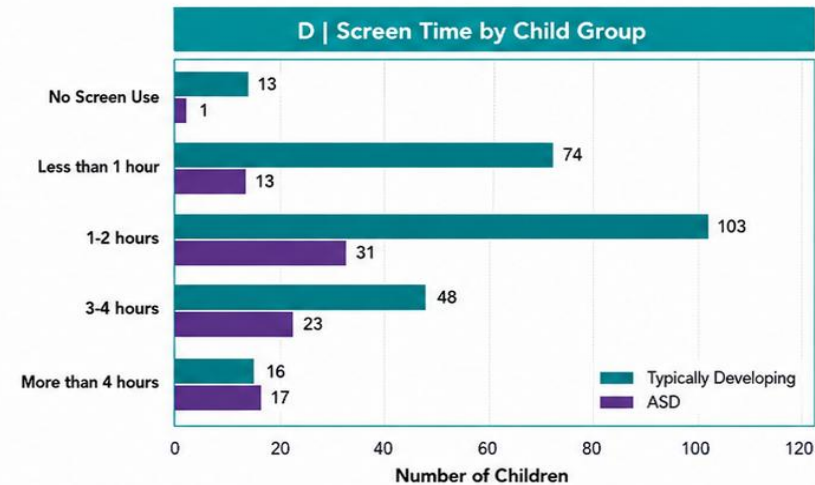
Distribution of total daily screen time across all surveyed children.



Most frequent times of day when children are exposed to screens.



Primary type of content consumed by the children.



Comparison of daily screen time across TD and ASD subgroups.

SCREEN EXPOSURE HIGHLIGHTS

RELIABILITY TEST



Small Summary Table

Scale	Items	Cronbach's α	Interpretation
Family Environment	5	0.693	Acceptable
Mental Development	8	0.875	Strong



Conclusion

Both composite scales were reliable and suitable for further statistical analysis.



Key Findings

- Reliability tested using **Cronbach's Alpha**
- Mental Development Scale showed **strong internal consistency**
- Family Environment Scale showed **acceptable reliability**
- All items contributed meaningfully to the scales
- No item deletion improved reliability significantly

NORMALITY TEST



Small Summary Table

Variable	N	Shapiro-Wilk Stat	Shapiro-Wilk p-value	K-S Stat	K-S p-value	Is Normal ?
Mental Development Score	339	0.9398	1.68e-10	0.1272	3.06e-05	No
Family Environment Score	339	0.9508	3.23e-09	0.1046	0.0011	No



Conclusion

Non-parametric tests were used:
Spearman Correlation
Mann-Whitney U Test
Kruskal-Wallis Test



Interpretation

Normality tested using: **Shapiro-Wilk Test**,
Kolmogorov-Smirnov Test, **Q-Q Plots**

All variables showed:

Significant p-values ($p < 0.05$)

Deviation from normal distribution

Q-Q plots confirmed **non-normality**



BIVARIATE AND MULTIVARIATE ANALYSIS

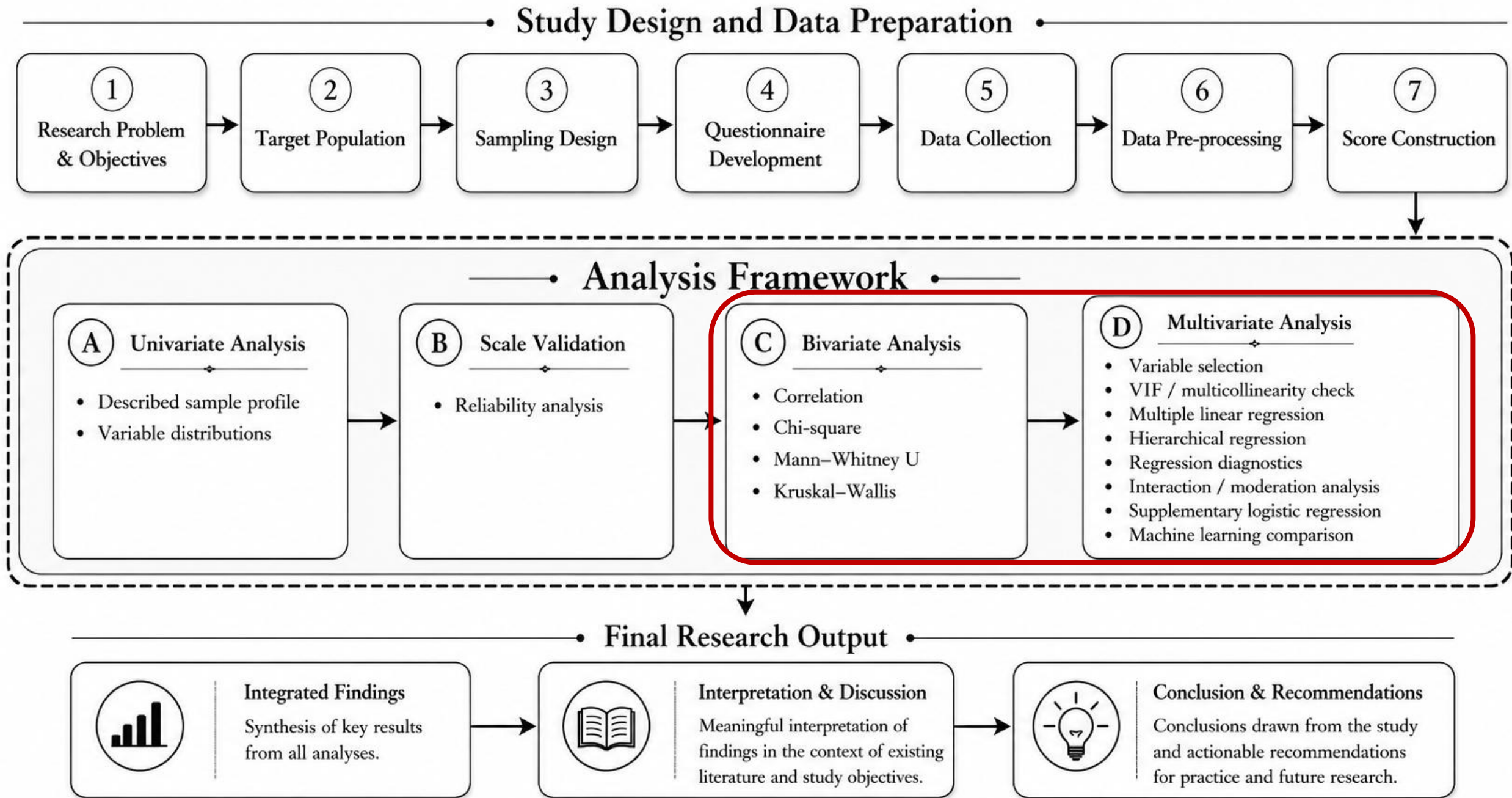


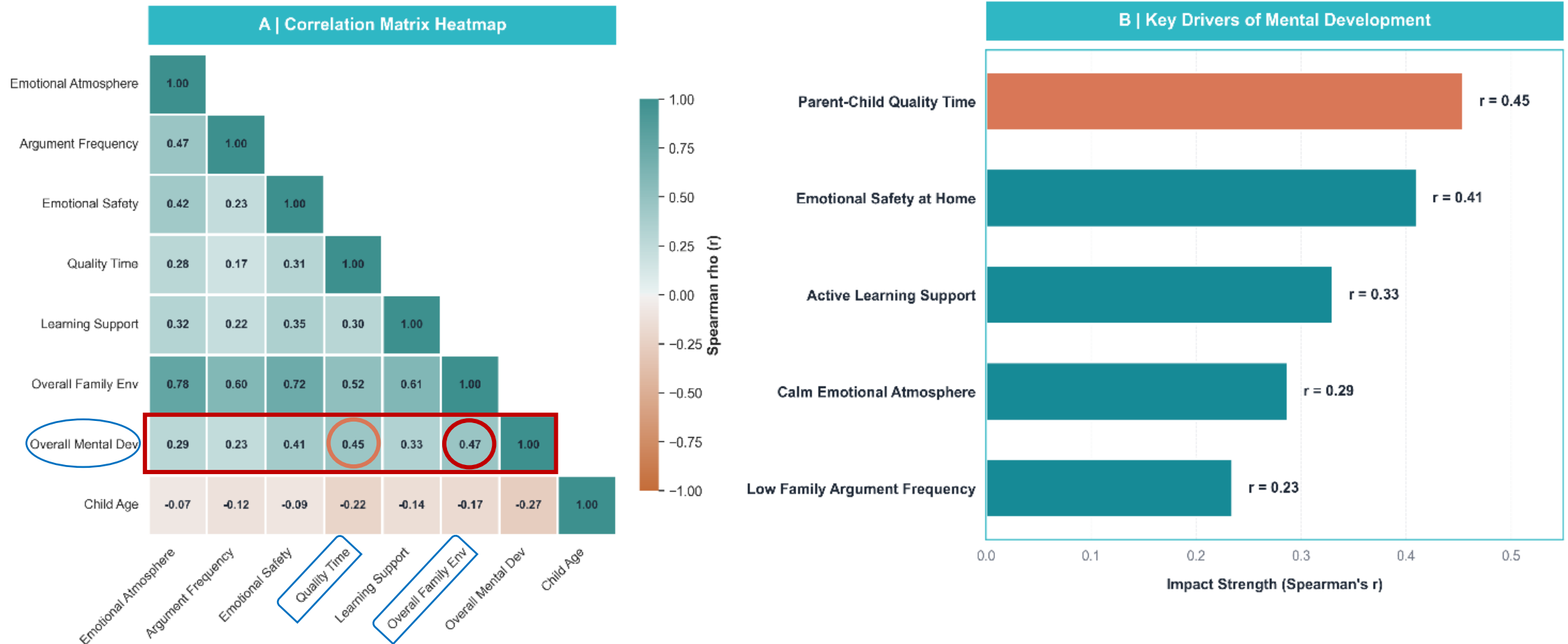
Figure: Overall Research Framework for the Study

SPEARMAN'S CORRELATION ANALYSIS

Q1: Is family environment associated with mental well-being?

Q2: Which family environment indicators are most related to children's mental well-being?

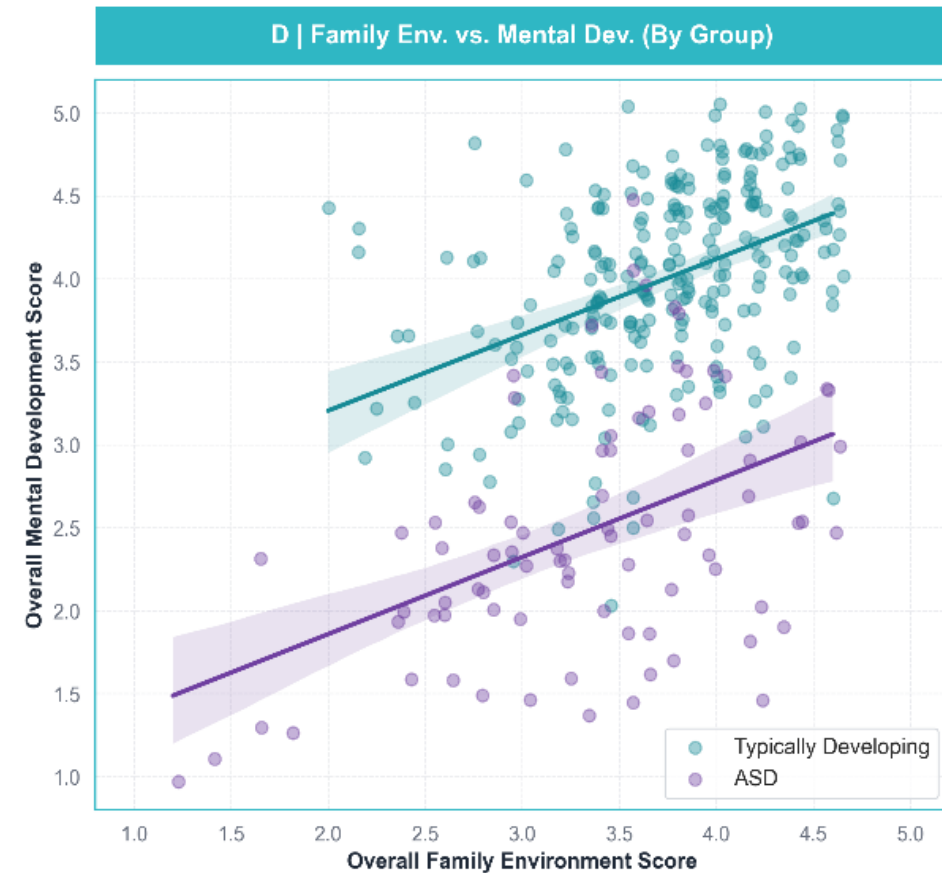
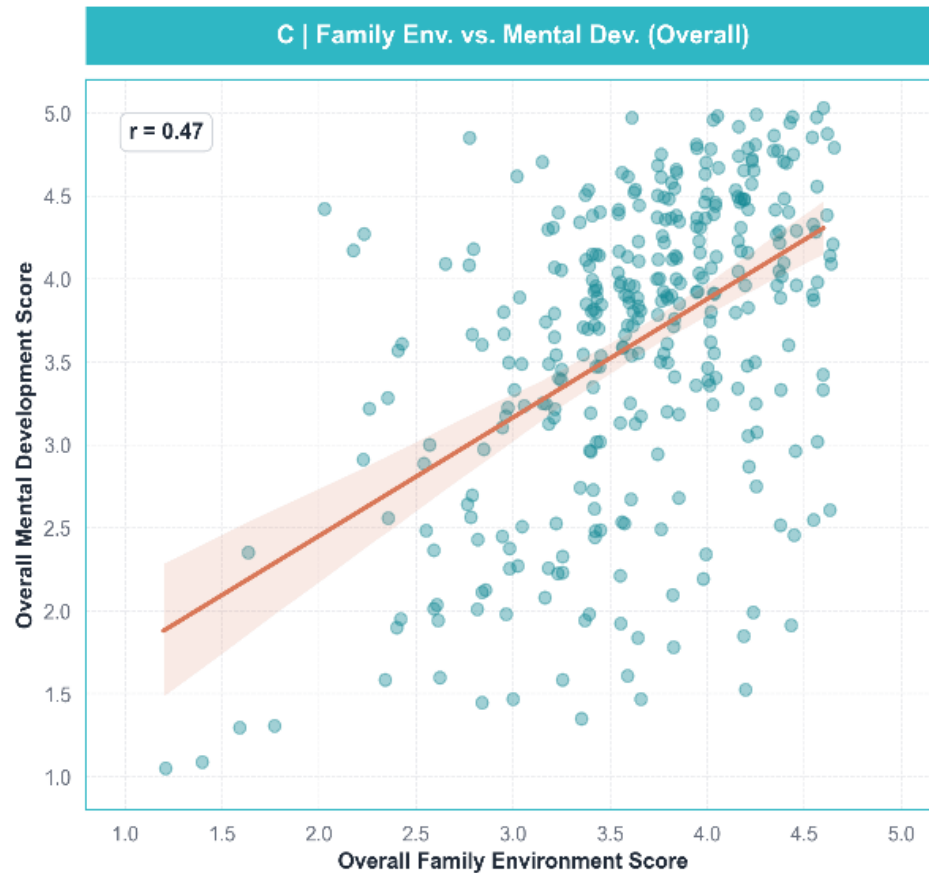
Family Environment & Child Mental Development Analysis



CORRELATION ANALYSIS

Q3: Do children with better family environments tend to have better mental well-being?

Q4: Do TD children generally show higher mental well-being than ASD children?



Note: Heatmap (Panel A) displays relationships across all variables. Key Drivers (Panel B) isolates individual family factors impacting mental development. Panels C & D illustrate the positive overall correlation and contrast differences between Typically Developing and ASD subgroups.

CHI-SQUARE TESTS OF INDEPENDENCE

Purpose : To examine whether categorical variables are associated with Child Group (ASD vs. TD). Effect size measured using Cramér's V.

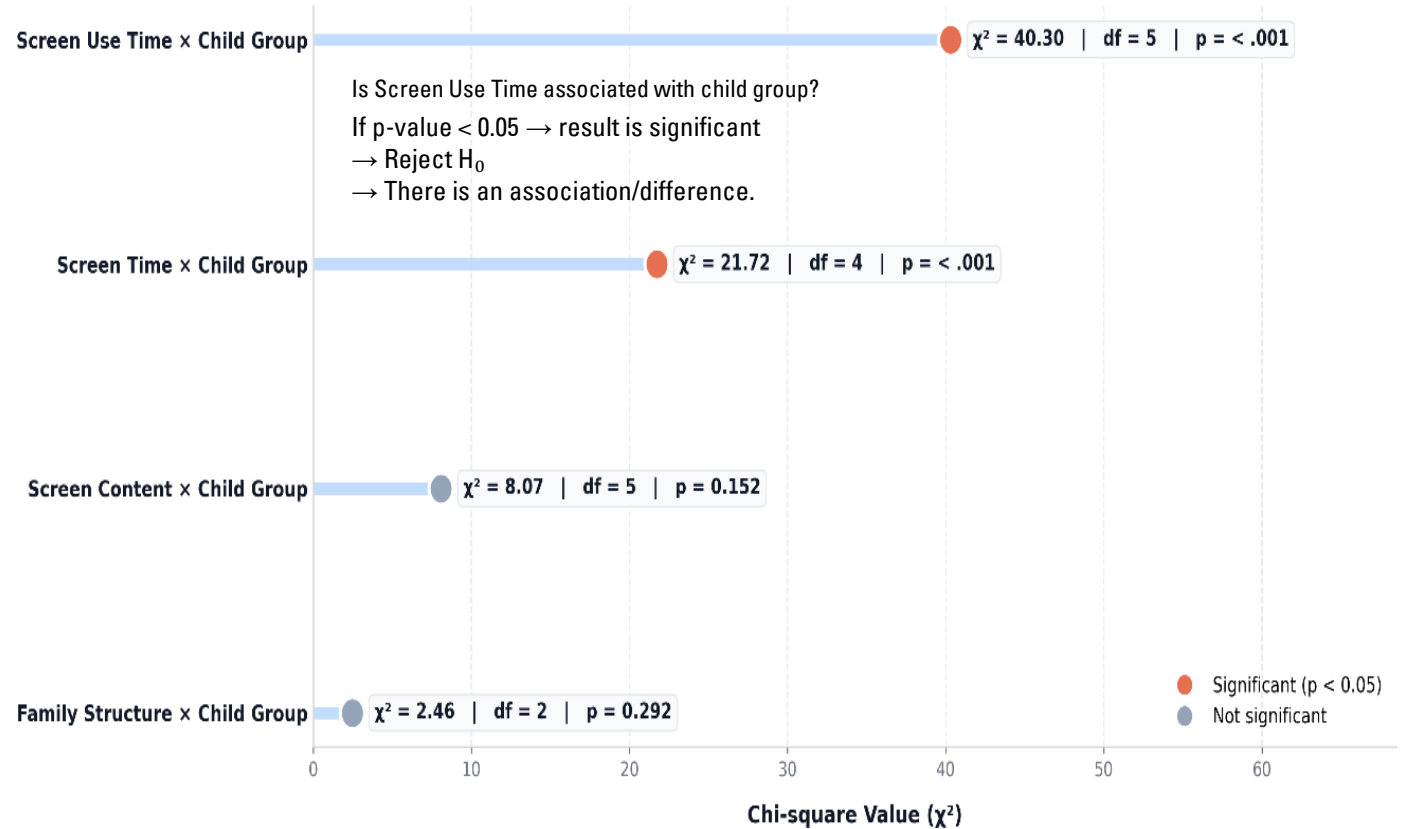
Hypothesis :

H0: There is no significant association

H1: There is a significant association

Variables	χ^2	df	p-value	Cramér's V	Effect Size	Significant
Family_Type × Child_Group	2.463	2	0.292	0.085	Very weak	No
Screen_Time × Child_Group	21.724	4	< .001	0.253	Weak	Yes
Screen_Content × Child_Group	8.074	5	0.152	0.154	Weak	No
Screen_Use_Time × Child_Group	40.299	5	< .001	0.345	Moderate	Yes

Chi-Square Test : Association with Child Groups



MANN–WHITNEY U TEST

Purpose : To compare a **numerical score** between **two independent groups**. The effect size is measured using the rank-biserial correlation (r).

Hypothesis

H_0 : There is no difference in family env scores between groups
 H_1 : There is a Significant difference in family env scores between groups

COMPARISON	SCORE	U	P-VALUE	R	EFFECT	SIGNIFICANT
TD vs. ASD	Mental WB	20,241.0	< .001	−0.875	Strong	Yes
Male vs. Female	Mental WB	12,535.0	0.388	0.056	Very weak	No
TD vs. ASD	Family Env	14,233.5	< .001	−0.319	Moderate	Yes

KRUSKAL WALLIS TEST

Purpose : To compare a **numerical score** between **three or more independent groups**. The effect size is measured using the Epsilon Squared ϵ^2 .

Hypothesis

H_0 : Scores do not differ across groups.

H_1 : Scores differ significantly across groups.

Grouping Variable	Score Variable	H	df	p-value	Epsilon Squared ϵ^2	Effect	Significant?
Family Type	Mental WB	3.245	2	0.197	0.004	Very weak	No
Family Type	Family Env	11.640	2	0.003	0.029	Weak	Yes
Mother Education	Mental WB	45.888	7	< .001	0.117	Moderate	Yes
Father Education	Mental WB	20.463	8	0.009	0.038	Weak	Yes
Father Occupation	Family Env	8.413	7	0.296	0.004	Very weak	No
Mother Occupation	Family Env	11.558	4	0.021	0.023	Weak	Yes
Screen Time	Mental WB	13.928	3	0.003	0.033	Weak	Yes

MULTIPLE REGRESSION

Key Question:
Does family environment still explain children’s mental well-being after controlling for family structure and other background factors?

Variable Group	Variables
Outcome Variable	Mental Well-Being Score
Selected Predictor Variables	Family Environment Score, Family Type, Child Group, Child Age, Child Gender, Household Size, Screen Time, Father’s Education, Mother’s Education, Monthly Income
Excluded Variables	Primary Caregiver, School Status, Screen Content, Screen Use Time, Father’s Occupation, Mother’s Occupation, Family Environment Item Scores, Mental Well-Being Item Scores

Variable Selection For Modelling

Model	Variables Added	Purpose
Model 1	Child factors	Controls basic child profile
Model 2	Model 1 + Family type + household size	Tests family structure effect
Model 3	Model 2+ Family Environment Score	Tests family environment effect
Model 4	Model 3 + Screen Time	Adds screen exposure
Model 5	Model 4 + Parental education + income	Final adjusted model

Model Plan

MULTIPLE REGRESSION

Hypothesis
H₀:The selected predictors do not significantly explain children’s Mental Well-Being Score.
H₁:At least one selected predictor significantly explains children’s Mental Well-Being Score.

Indicator	Value
N	339
R ²	0.685
Adjusted R ²	0.652
F-statistic	20.753
Model p-value	< .001

Summary Model Fit

Interpretation: The overall regression model was statistically significant, $p < .001$. This means the at least one selected predictor significantly explains children’s Mental Well-Being Score.

Hypothesis
H₀:The predictor has no significant effect on Mental Well-Being Score.
H₁:The predictor has a significant effect on Mental Well-Being Score.

Predictor	B	p-value
Family Environment Score	0.414	< .001
Child Group: ASD	−1.204	< .001
Mother Education: HSC	−0.271	0.012
Mother Education: No schooling	−0.623	0.006
Mother Education: Primary	−0.675	< .001
Mother Education: SSC	−0.295	0.013

Significant Adjusted Predictors

Interpretation: Family environment, ASD status, and mother’s education were the key adjusted predictors of children’s mental well-being.

Hierarchical Model Comparison						
Model	Block Added	R ²	Adjusted R ²	ΔR ²	p-value for Change	Decision
Model 1	Child factors	0.5425	0.5384	Baseline	Baseline	Baseline model
Model 2	Family structure and household context	0.5452	0.5370	0.0027	0.5848	No significant improvement
Model 3		0.6291	0.6212	0.0839	< .001	Significant improvement
Model 4	Screen exposure	0.6455	0.6335	0.0164	0.0051	Significant improvement
Model 5	Socioeconomic background	0.6846	0.6516	0.0391	0.0175	Significant improvement

BINARY LOGISTIC REGRESSION

Purpose : To identify factors related to high mental well-being among children.

Mental Well-Being Score was converted into two groups using the median cutoff (3.875)

Mental Well-Being Group	Criteria	n	%
High Mental Well-Being	Score $\geq$ 3.875	173	51.0%
Low Mental Well-Being	Score $<$ 3.875	166	49.0%

H₀:Selected factors are not significantly associated with high mental well-being.
H₁:At least one selected factor is significantly associated with high mental well-being.

Model	Outcome	N	Log-Likelihood	Null Log-Likelihood	McFadden Pseudo R ²	LLR p-value	AIC	BIC
Adjusted Binary Logistic Regression	High Mental Well-Being vs Low Mental Well-Being	339	-140.495	-234.905	0.4019	< .001	346.991	473.249

BINARY LOGISTIC REGRESSION

Significant Predictors of High Mental Well-Being

Predictor	Adjusted OR	95% CI	p-value	Interpretation
Family Environment Score	5.8638	3.0461–11.2881	< .001	Better family environment increased the odds of high mental well-being.
Child Group: ASD	0.0219	0.0066–0.0722	< .001	ASD children had much lower odds of high mental well-being than TD children.
Screen Time: 1–2 hours	2.4013	1.0660–5.4089	0.0345	Children with 1–2 hours of screen time had higher odds of high mental well-being than the reference group.
Mother Education: Primary	0.0624	0.0087–0.4458	0.0057	Children whose mothers had primary education had lower odds of high mental well-being than those whose mothers had Master’s degree or higher.

Model Evaluation

Accuracy	Precision	Recall	F1-score	ROC-AUC
0.8201	0.7947	0.8728	0.8320	0.8931

ML MODEL COMPARISON

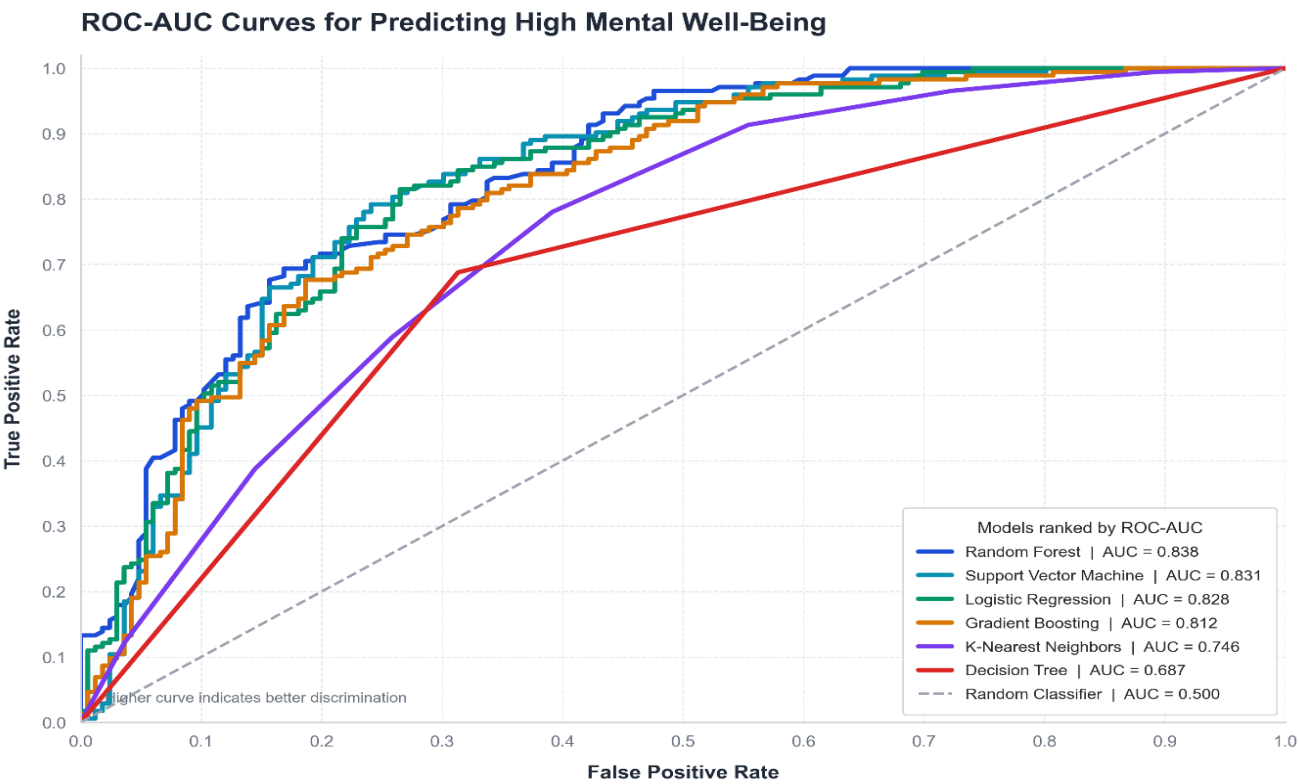
How It Was Done : Used 5-fold stratified cross-validation

Key Message

Random Forest achieved the highest ROC-AUC, **while Logistic Regression** gave the best accuracy and F1-score.

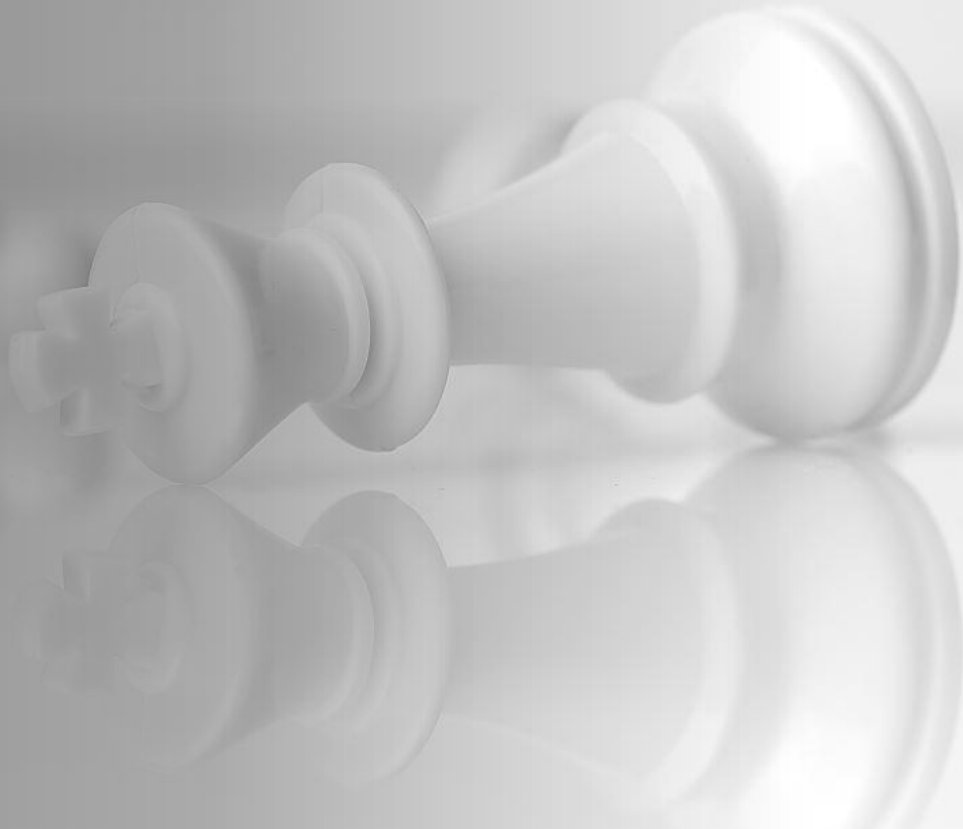
Machine learning supported the main findings, but regression remained the main explanatory model.

Model	Accuracy	Precision	Recall	F1-score	ROC-AUC
Random Forest	0.7286	0.7263	0.7576	0.7402	0.8400
Logistic Regression	0.7727	0.7562	0.8207	0.7868	0.8324
Support Vector Machine	0.7699	0.7511	0.8271	0.7862	0.8296
Gradient Boosting	0.7285	0.7256	0.7573	0.7402	0.8146
K-Nearest Neighbors	0.6961	0.6766	0.7810	0.7233	0.7441
Decision Tree	0.6874	0.6997	0.6879	0.6925	0.6875



Compared 6 classification models

CONCLUSION & RECOMMENDATION



KEY TAKEAWAY MESSAGES

01

**Family Environment
Matters Most**

Quality of care is more important than family type.

02

**ASD Children Need
Extra Support**

ASD children showed lower mental well-being than Typically Developing children.

03

**Mother's Education Is
Important**

Higher maternal education was linked with better child outcomes.

04

**Positive Home
Environment Makes a
Difference**

A calm, caring, and supportive home helps children develop better mental well-being.

RECOMMENDATION



For Parents: Build a calm, safe, and supportive home.



For Policy: Promote parenting education and awareness.



For Schools: Give extra care and support to ASD children.



For Community: Encourage Healthy and balanced screen habits

STRENGTHS & LIMITATIONS

Strengths

- Good sample size: Included 339 children
- Compared typically developing and ASD children
- Used multiple analyses: descriptive, bivariate, multivariate, and ML

Limitations

- Cross-sectional study
 - Non-random sampling (purposive sampling)
 - ASD and single-parent groups were smaller.
 - Parent-reported data
 - Urban Pabna only
-

CONCLUSION

“It takes a village to raise a child.” — African Proverb

But our study reminds us that the first village of every child is the family — where care, safety, support, and love shape their mental well-being.

THANK YOU & Q&A



Presented by:

Sumaiya, Hanif, Shakil, Kabbo, Monisha



Supervisor:

Dr. Md. Shamim Reza

Professor, Department of Statistics
Pabna University of Science and Technology



We welcome your questions.

